

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 6: Security, Privacy and Data Integrity

Subtopic 6.1: Data Security

Concept 6.1.3: Network and Internet Threats (Malware, Hacking, Phishing)

Total Questions: 6

Mark Schemes begin on Page 4

Question 1 | Source: 9618_w25_qp_11 (Q8.a)

8 One ethical consideration for a student connecting their personal computer to the school network is the risk of spreading malware on the network.

(a) Viruses and pharming are examples of malware.

Explain what is meant by a virus and pharming.

Virus
.....
.....

Pharming
.....
.....

[2]

Question 2 | Source: 9618_w23_qp_12 (Q5.c)

(c) Describe the following threats to a computer system.

Phishing email
.....
.....

Spyware
.....
.....

[4]

Question 3 | Source: 9618_w23_qp_13 (Q8.b)

(b) State **one** difference and **one** similarity between pharming and phishing.

Difference
.....

Similarity
.....

[2]

Question 4 | Source: 9618_w23_qp_13 (Q8.c)

(c) Explain how the data security risks of malware can be restricted.

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..... [3]

Question 5 | Source: 9618_w21_qp_11 (Q2.c)

2 Xanthe wants to maintain the integrity and security of data stored on her computer.

(c) Two malware threats are spyware and viruses.

Give **two** similarities and **one** difference between spyware and a virus.

Similarity 1

.....

Similarity 2

.....

Difference

..... [3]

Question 6 | Source: 9618_s21_qp_12 (Q8.c)

8 A company has several security measures in place to prevent unauthorised access to the data on its computers.

(c) The company is also concerned about threats posed by networks and the internet.

Identify **two** threats to the data that are posed by networks and the internet.

Threat 1

.....

Threat 2

..... [2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q8.a)

8(a)	1 mark for each correct explanation, max 2 marks Virus: Malicious program/software that replicates/copies itself and deletes/alters files/data stored on a computer Pharming: Malicious code/software installed on a computer which redirects user to a fake website to obtain personal data	2
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Mark Scheme for Question 2 | Source: 9618_w23_ms_12 (Q5.c)

5(c)	1 mark for each bullet point. Phishing email (max 2) - The email pretends to be from an official body ... persuading individuals to disclose private information // by example such as bank details ... or requesting authentication by redirecting to an unofficial/unauthorised website // inviting a user to click a link Spyware (max 2) - Malware downloaded without the user's knowledge ... which secretly records the user's actions / keystrokes on the computer ... and sends logs of the actions to a third party	4
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Mark Scheme for Question 3 | Source: 9618_w23_ms_13 (Q8.b)

8(b)	1 mark for difference 1 mark for similarity Difference: - Pharming is malicious code that redirects to a fake website. Phishing uses an email to prompt user action. - Pharming is automatic. Phishing requires user action. Similarity: - Both try to obtain financial or personal information - Both are a false representation of an official organisation, e.g. a bank - Both make use of fake websites	2
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Mark Scheme for Question 4 | Source: 9618_w23_ms_13 (Q8.c)

8(c)	1 mark for each bullet point (max 3). • Download programs from reputable websites / sources • ...as these are less likely to contain malware • Backup / archive computer systems • ...so they can be restored in case of data loss from malware program installation • Install and run anti-malware program • ...so that regular scans can be made for known malware • ...and if malware is found it can be quarantined / removed • ...and computer's anti-malware definitions are regularly updated • Using a firewall to block unused ports • ...so that malware cannot enter the computer system • Deny administrator privileges to everyday users • ...so that malware cannot be downloaded by everyday users • Avoid the use of / access to removable devices • ...so that malware cannot be installed from these devices	3
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Mark Scheme for Question 5 | Source: 9618_w21_ms_11 (Q2.c)

2(c)	1 mark per similarity to max 2 • Both are pieces of malicious software • Both are downloaded / installed/run without the user's knowledge • Both can pretend to be / are embedded in other legitimate software when downloaded // both try to avoid the firewall • Both run in the background 1 mark for difference • Virus can damage computer data; spyware only records / accesses data • Virus does not send data out of the computer; spyware sends recorded data to third party • Virus replicates itself; spyware does not replicate itself	3
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Mark Scheme for Question 6 | Source: 9618_s21_ms_12 (Q8.c)

8(c)	1 mark per correct answer to max 2 • Malware // viruses // spyware // by example • Hacking • Phishing • Pharming	2
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